

MedForge Workbook — Acromegaly

Educational draft - source review required

Big picture

Acromegaly is a chronic endocrine disorder caused by hypersecretion of growth hormone (GH), most commonly due to a pituitary adenoma, leading to elevated insulin-like growth factor 1 (IGF-1) levels [S1]. It presents insidiously with progressive physical changes and systemic complications, including metabolic disturbances, cardiovascular disease, and increased cancer risk [S4]. Early diagnosis and treatment are critical due to high morbidity and mortality, especially in older patients [S1][S5]. The disease is often missed for years, contributing to poor outcomes [S2].

Core mechanism

GH hypersecretion drives increased IGF-1 production, which mediates many of the disease's effects. GH directly impairs insulin signaling in peripheral tissues, promotes lipolysis, and reduces glucose uptake, leading to insulin resistance independent of obesity [S3]. This results in a lean phenotype with reduced adipose tissue mass despite insulin resistance [S3]. The pathophysiology creates a unique metabolic state where diabetes mellitus is common, even in the absence of obesity [S3].

Anatomy/physiology/pathology/pharmacology connections when supported

The pituitary gland is the primary source of GH overproduction, typically via a pituitary adenoma [S1][S4]. Elevated GH and IGF-1 levels are central to diagnosis and monitoring [S4]. Biochemical confirmation requires elevated serum IGF-1 and lack of GH suppression after oral glucose administration [S4]. Imaging, particularly pituitary MRI, is advised to identify the underlying adenoma [S4].

Treatment options include transsphenoidal surgery (first-line for accessible tumors), somatostatin receptor ligands (e.g., octreotide, lanreotide), cabergoline, and pegvisomant (a GH receptor antagonist) [S4]. Surgery is effective in many but fails to achieve remission in patients with large or invasive macroadenomas [S4].

Diabetes mellitus is a common complication due to GH-induced insulin resistance, and patients have increased cardiovascular risk [S3].

Exam traps

- Confusing acromegaly with gigantism: gigantism occurs in children before epiphyseal closure, while acromegaly develops in adults after growth plates fuse [S1].
- Assuming all patients with acromegaly have diabetes: while ~1/3 develop diabetes, it is not universal [S3].
- Believing acromegaly is rare: it is rare but incidence may be rising in the elderly due to increased life expectancy [S5].
- Overestimating cancer risk: while long-term GH exposure may promote tumor development, epidemiological evidence remains inconclusive, and the association is debated [S6].

Common confusions

- Acromegaly is not caused by excess GH alone; it is defined by hypersecretion of GH and elevated IGF-1 [S1].
- GH-induced insulin resistance occurs independently of obesity and is paradoxically linked to low body fat [S3].

- The disease is not limited to young adults: elderly patients are increasingly diagnosed, and comorbidities are more prevalent in this group [S5].
- Treatment success is not guaranteed: even after surgery, many patients with macroadenomas remain non-remitted [S4].

10 retrieval questions

1. What is the primary cause of acromegaly? [S1][S4]
2. Which metabolic complication is most commonly associated with acromegaly and occurs in approximately one-third of patients? [S3]
3. How does growth hormone impair insulin signaling in acromegaly? [S3]
4. What imaging modality is recommended for identifying the underlying pituitary adenoma in acromegaly? [S4]
5. What is the first-line treatment for acromegaly in patients with accessible pituitary adenomas? [S4]
6. Why is early diagnosis crucial in acromegaly? [S1][S4]
7. What is the typical age range for diagnosis of acromegaly in younger patients? [S5]
8. What are the independent predictors of mortality in acromegaly according to current evidence? [S6]
9. How does the risk of diabetes mellitus in acromegaly differ from that in typical type 2 diabetes? [S3]
10. What is the current status of evidence regarding the link between acromegaly and cancer development? [S6]

Five-minute recap

Acromegaly results from GH hypersecretion due to pituitary adenoma, leading to elevated IGF-1 and systemic complications [S1][S4]. GH directly causes insulin resistance and metabolic dysfunction, even in lean patients [S3]. Diagnosis relies on biochemical testing and MRI [S4]. Treatment begins with surgery, followed by medical therapy if needed [S4]. Diabetes is common and carries cardiovascular risk [S3]. Mortality is linked to age, GH levels, and IGF-1 at diagnosis [S6]. The disease is increasingly seen in the elderly, and delayed diagnosis remains a key challenge [S5].

Educational study aid — verify important claims against the cited sources.

Practice Questions

Question: Acromegaly is primarily caused by hypersecretion of which hormone? [S1] [S3] [S4]

- A) Adrenocorticotrophic hormone
- B) Thyroid-stimulating hormone
- C) Growth hormone
- D) Prolactin

Correct Answer: C) [S1] [S3] [S4]

Rationale: Growth hormone hypersecretion is the defining feature of acromegaly, leading to elevated insulin-like growth factor 1. [S1] [S3] [S4]

Question: Which laboratory test is used to confirm the diagnosis of acromegaly after glucose administration? [S4]

- A) Serum cortisol
- B) Serum insulin-like growth factor 1 (IGF-1)
- C) Lack of growth hormone suppression after glucose administration
- D) Serum TSH level

Correct Answer: C) [S4]

Rationale: The diagnosis of acromegaly is confirmed biochemically by lack of growth hormone suppression after glucose administration. [S4]

Question: What is the most common cause of acromegaly? [S3] [S4]

- A) Adrenal adenoma
- B) Pituitary adenoma
- C) Thyroid carcinoma
- D) Pancreatic tumor

Correct Answer: B) [S3] [S4]

Rationale: Acromegaly most commonly arises due to a pituitary adenoma. [S3] [S4]

Question: Which comorbidity is commonly seen in patients with acromegaly and is driven by growth hormone-induced insulin resistance? [S3]

- A) Hypothyroidism
- B) Diabetes mellitus
- C) Hyperuricemia
- D) Hyperlipidemia

Correct Answer: B) [S3]

Rationale: Diabetes mellitus occurs in approximately one-third of acromegaly patients due to growth hormone-induced attenuation of insulin signaling. [S3]

Question: What is a unique metabolic feature of diabetes mellitus in acromegaly? [S3]

- A) Obesity is always present
- B) Insulin resistance occurs independently of obesity
- C) Glucose uptake increases in adipose tissue
- D) Insulin resistance is due to diet

Correct Answer: B) [S3]

Rationale: Insulin resistance in acromegaly occurs independently of obesity and is associated with a lean phenotype and reduced body adipose tissue mass. [S3]

Question: Which of the following is a recognized long-term complication of acromegaly? [S4]

- A) Hypothyroidism
- B) Cardiovascular disease
- C) Hypogonadism
- D) Osteoporosis

Correct Answer: B) [S4]

Rationale: Acromegaly is associated with cardiovascular, respiratory, endocrine, metabolic, musculoskeletal, and neoplastic comorbidities. [S4]

Question: What is the first-line treatment for acromegaly? [S4]

- A) Radiation therapy
- B) Somatostatin receptor ligands
- C) Transsphenoidal pituitary surgery
- D) Cabergoline monotherapy

Correct Answer: C) [S4]

Rationale: Transsphenoidal pituitary surgery is generally first-line therapy for patients with acromegaly. [S4]

Question: Which patients are less likely to achieve remission after pituitary surgery for acromegaly? [S4]

- A) Patients with small, non-invasive tumors
- B) Patients with macroadenomas
- C) Patients under 30 years old
- D) Patients with low IGF-1 levels

Correct Answer: B) [S4]

Rationale: Patients with larger and invasive tumors (macroadenomas) are often not in remission postoperatively. [S4]

Question: Which medication is considered a medical therapy for acromegaly? [S4]

- A) Insulin
- B) Pegvisomant
- C) Levothyroxine
- D) Spironolactone

Correct Answer: B) [S4]

Rationale: Medical therapies for acromegaly include somatostatin receptor ligands, cabergoline, and pegvisomant. [S4]

Question: What is a key factor associated with increased mortality in acromegaly? [S6]

- A) Low IGF-1 levels at diagnosis
- B) High IGF-1 levels at diagnosis
- C) Normal growth hormone levels
- D) Absence of comorbidities

Correct Answer: B) [S6]

Rationale: Higher insulin-like growth factor-1 (IGF-1) levels at diagnosis are an independent predictor of mortality in acromegaly. [S6]

Question: Which group is increasingly being diagnosed with acromegaly due to rising life expectancy? [S5]

- A) Children under 10
- B) Middle-aged adults
- C) Elderly patients
- D) Pregnant women

Correct Answer: C) [S5]

Rationale: Recent studies show an increase in the incidence and prevalence of acromegaly in the elderly, likely due to increasing life expectancy. [S5]

Question: What is a major challenge in diagnosing acromegaly? [S1] [S2]

- A) Rapid progression of symptoms
- B) Delayed diagnosis due to insidious onset
- C) Lack of laboratory tests
- D) Absence of clinical symptoms

Correct Answer: B) [S1] [S2]

Rationale: Acromegaly is an insidious disease, often missed for years, resulting in significantly delayed diagnosis. [S1] [S2]

Question: What is the relationship between growth hormone and insulin signaling in acromegaly? [S3]

- A) Growth hormone enhances insulin signaling
- B) Growth hormone stimulates insulin production
- C) Growth hormone directly attenuates insulin signaling
- D) Growth hormone has no effect on insulin

Correct Answer: C) [S3]

Rationale: Increased exposure to growth hormone directly attenuates insulin signaling, leading to decreased glucose uptake in peripheral tissues. [S3]

Question: What is a proposed predictor of mortality in acromegaly? [S6]

- A) Young age at diagnosis
- B) Low growth hormone levels
- C) Older age and higher GH levels at last follow-up
- D) Normal IGF-1 levels

Correct Answer: C) [S6]

Rationale: Older age, increased growth hormone levels at last follow-up, and higher IGF-1 levels at diagnosis are independent predictors of mortality. [S6]

Question: What is a potential long-term consequence of chronic growth hormone exposure in acromegaly? [S6]

- A) Increased risk of benign or malignant tumors
- B) Decreased risk of cancer
- C) Improved immune function

D) Reduced cardiovascular risk

Correct Answer: A) [S6]

Rationale: Long-term exposure to growth factors may lead to complications such as the development of benign or malignant tumors. [S6]

Question: What is the recommended approach for surveillance in patients with acromegaly? [S6]

A) Annual physical exams only

B) Rigorous surveillance of endocrine diseases

C) Routine cancer screening without personalization

D) No surveillance needed

Correct Answer: B) [S6]

Rationale: Rigorous surveillance of endocrine diseases may contribute to increased tumor detection. [S6]

Question: How does the prognosis of acromegaly correlate with patient age? [S5]

A) It improves with increasing age

B) It correlates inversely with age

C) It is unaffected by age

D) It improves with younger age

Correct Answer: B) [S5]

Rationale: The prognosis correlates inversely with the patient's age, disease duration, and last GH level. [S5]

Question: What is a limitation in current research regarding diabetes mellitus in acromegaly? [S3]

A) Lack of awareness among clinicians

B) Limited research on management strategies

C) Absence of diagnostic tools

D) Inadequate patient follow-up

Correct Answer: B) [S3]

Rationale: Specific recommendations for the management of diabetes mellitus secondary to acromegaly are lacking due to limited research on this subject. [S3]

Question: Which therapy is generally effective as both primary and postoperative treatment in acromegaly? [S5]

A) Radiation therapy

B) Somatostatin receptor ligands

C) Pegvisomant

D) Cabergoline

Correct Answer: B) [S5]

Rationale: Somatostatin receptor ligands are generally effective as both primary and postoperative treatment. [S5]

Question: What is the primary reason for delayed diagnosis of acromegaly in elderly patients? [S5]

- A) Lack of symptoms
- B) Similarities between aging and disease characteristics
- C) Inadequate screening
- D) Poor patient awareness

Correct Answer: B) [S5]

Rationale: The delay in diagnosis is aggravated by the similarities of the aging process with some of the characteristics of the disease. [S5]

Viva

1. What is the primary cause of acromegaly?

Hypersecretion of growth hormone, most commonly due to a pituitary adenoma [S3][S4].

2. What is the key biochemical marker used to confirm acromegaly?

Elevated serum insulin-like growth factor 1 (IGF-1) with lack of growth hormone suppression after glucose administration [S1][S4].

3. How does growth hormone contribute to insulin resistance in acromegaly?

Growth hormone directly attenuates insulin signaling and stimulates lipolysis, reducing glucose uptake in peripheral tissues [S3].

4. Why is diabetes mellitus common in patients with acromegaly?

Insulin resistance occurs independently of obesity and is linked to growth hormone exposure, even in lean patients [S3].

5. What is the typical age of onset for acromegaly?

Most commonly diagnosed during the 3rd to 4th decade of life [S5].

6. Why is acromegaly more frequently diagnosed in the elderly today?

Due to increased life expectancy, and because aging symptoms overlap with acromegaly manifestations [S5].

7. What is the first-line treatment for acromegaly?

Transsphenoidal pituitary surgery [S4][S5].

8. What comorbidities are associated with acromegaly?

Cardiovascular, respiratory, endocrine, metabolic, musculoskeletal, and neoplastic complications [S4].

9. What are the main medical therapies used for acromegaly?

Somatostatin receptor ligands, cabergoline, and pegvisomant [S4].

10. How does acromegaly affect mortality?

Increased morbidity and mortality are associated with the disease, especially with prolonged disease duration and high GH levels [S1][S6].

11. What factors are independent predictors of mortality in acromegaly?

Older age, higher GH levels at last follow-up, higher IGF-1 at diagnosis, malignancy, and radiotherapy [S6].

12. Is there a proven link between acromegaly and cancer development?

The association remains debated due to limited epidemiological data; long-term growth factor exposure may increase tumor risk, but evidence is inconclusive [S6].

13. What imaging study is recommended in acromegaly patients?

Pituitary magnetic resonance imaging to identify underlying pituitary adenoma [S4].

14. Why is early diagnosis crucial in acromegaly?

Delayed diagnosis is common, and early treatment reduces excess mortality and complications [S1][S4].

15. What is the significance of diabetes mellitus in acromegaly patients?

It is associated with increased cardiovascular morbidity and mortality, and specific management recommendations are currently lacking [S3].

Case Exercises

Mini-Case 1: Delayed Diagnosis in an Elderly Patient

Question: A 72-year-old man presents with joint pain, thickened fingers, and increased sweating. His medical history includes hypertension and type 2 diabetes. Laboratory tests show elevated serum IGF-1 and lack of growth hormone suppression after glucose tolerance testing. What is the most likely diagnosis, and why is diagnosis delayed in this age group? [S5]

Teaching Answer: The most likely diagnosis is acromegaly. Diagnosis is often delayed in elderly patients due to the overlap between aging-related symptoms and those of acromegaly, such as joint pain and sweating, and the insidious progression of the disease. [S5]

Discussion: Acromegaly is typically diagnosed in the third to fourth decades, but incidence increases in the elderly due to longer life expectancy. The clinical presentation mimics aging, leading to delayed recognition. [S5]

Mini-Case 2: Diabetes Mellitus in Acromegaly

Question: A 55-year-old woman with acromegaly develops hyperglycemia and insulin resistance despite no significant weight gain. What pathophysiological mechanism explains the development of diabetes mellitus in acromegaly, and how does it differ from obesity-related insulin resistance? [S3]

Teaching Answer: Growth hormone directly attenuates insulin signaling and stimulates lipolysis, leading to decreased glucose uptake in peripheral tissues. This insulin resistance occurs independently of obesity and is paradoxically associated with a lean phenotype and reduced body adipose tissue mass. [S3]

Discussion: In acromegaly, insulin resistance is driven by growth hormone exposure, not adiposity. This makes it a unique model of insulin resistance. [S3]

Mini-Case 3: Treatment Options for Acromegaly

Question: A 48-year-old man with a macroadenoma is being evaluated for treatment of acromegaly. After transsphenoidal surgery, his growth hormone levels remain elevated. What is the next-line medical therapy, and what is its mechanism of action? [S4]

Teaching Answer: Somatostatin receptor ligands, such as octreotide or lanreotide, are effective as both primary and postoperative treatment. These agents bind to somatostatin receptors on pituitary cells, inhibiting growth hormone secretion. [S4]

Discussion: For patients with macroadenomas who fail to achieve remission after surgery, medical therapy is essential to control growth hormone levels. [S4]

Mini-Case 4: Cardiovascular Risk in Acromegaly

Question: A patient with acromegaly has a history of hypertension and elevated lipid levels. Which comorbidity is most strongly associated with increased cardiovascular morbidity and mortality in acromegaly, and why? [S3]

Teaching Answer: Diabetes mellitus is associated with increased cardiovascular morbidity and mortality in acromegaly due to growth hormone-induced insulin resistance and metabolic dysregulation. [S3]

Discussion: The combination of hyperglycemia, dyslipidemia, and insulin resistance increases cardiovascular risk, making diabetes a key comorbidity in acromegaly. [S3]

Mini-Case 5: Mortality and Disease Control

Question: A 60-year-old patient with acromegaly has a high IGF-1 level at diagnosis and elevated growth hormone at last follow-up. How do these factors relate to mortality risk in acromegaly, and what does this imply about disease management? [S6]

Teaching Answer: Higher IGF-1 at diagnosis and elevated growth hormone at last follow-up are independent predictors of mortality in acromegaly. This underscores the importance of early diagnosis and sustained control of growth hormone levels. [S6]

Discussion: Mortality improves with disease control and management of comorbidities, highlighting the need for rigorous monitoring and timely intervention. [S6]

References

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